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Interdependence between Urban Processes and Energy Transitions: The Dimensions of Urban Energy Transitions (DUET) Framework

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ABSTRACT

This paper presents a framework to study the interdependence between urban processes and energy transitions – the Dimensions of Urban Energy Transitions (DUET) framework. The framework emerges from current debates about the need to incorporate systematically the spatial and political dimensions of urban energy transitions. We seek to develop a systematic framework that brings together well-established principles of transitions theory with insights from recent debates about how transitions unfold in urban contexts. Illustrated through a comparative meta-analysis of 29 case studies of urban energy transitions, the DUET framework provides an approach that considers simultaneously three dimensions of energy transitions: socio-technical experimentation, urban political processes, and socio-spatial (re)configuration. The three dimensions interact closely with each other and together enforce 'transitions of cities'. From DUET framework, it argues that alignment between industry interests and territorial priorities is key to meaningful energy transitions.

1. Introduction

The field of transitions captures the co-dynamics of energy related technologies, institutions, and social and economic sub-systems (Turnheim et al., 2015). A body of critique has emerged concerned with the need to incorporate the complexity of analyses of space in processes leading to transitions to sustainability (Berkhout et al., 2009; Coenen et al., 2012; Truffer and Coenen, 2012; Bridge et al., 2013; Bergek et al., 2015; Hansen and Coenen, 2015). The field of transition studies has advanced contemporary understandings of the dynamics of socio-technical change over time (the temporal concept of 'transitions') for a given geographical unit, with a predominant focus on national-level dynamics (Hodson and Marvin, 2010a). However, a more contextualized and nuanced reading of spatiality is still lacking (Schwanen, 2017). Space has received less attention in sustainability transitions than other conditioning factors such as the technological fit to specific uses or the institutional mechanisms of governance that shape the adoption of social and technical innovations. For instance, existing theoretical approaches have largely played down the role of the heterogeneous socio-spatial characteristics within or across places, even though such characteristics are determinant features of transitions to sustainability (Bridge et al., 2013).

The tendency in socio-technical transitions studies to adopt an 'average' viewpoint for heterogeneous spatial contexts limits their capacity to explain urban sustainability transitions (Goess et al., 2015; Hu et al., 2015). There is a direct link between relations of distance and contiguity and innovation (Raven et al., 2012; Rutherford and Coutard, 2014; Wieczorek et al., 2015). Empirical studies have demonstrated that spatial concerns add analytical richness to existing frameworks (Binz et al., 2012; De Laurentis, 2013; Gosens

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et al., 2015). As Bridge et al. (2013) have argued, the "energy transition is fundamentally a geographical process that involves reconfiguring current spatial patterns of economic and social activity" (p. 331).

Spatial aspects are always visible in energy transitions in urban areas (Van den Bergh et al., 2011). Urban transformations are central for energy transitions as much as energy transitions shape the urban fabric (Rutherford and Coutard, 2014). Implementing new technologies will require changes to the urban fabric, adaptations within the built environment, changes in practices and redefinition of the local governance context. Furthermore, cities are increasingly regarded as facilitating the action of local 'managers' of energy transitions. Such local managers may be local governments, but also a wide range of other public and private actors such as companies, para-state institutions, international networks, community groups or NGOs (Bulkeley et al., 2014). Cities offer possibilities to foster new forms of social and technological innovation, and thus, they are closely linked to the development of transition experiments (Hodson and Marvin, 2010a; Evans and Karvonen, 2011; Evans et al., 2016; Voytenko et al., 2016). The idea of cities as 'urban laboratories' reflects the role that urban processes play in channeling actors, data and resources and fostering innovation (Evans and Karvonen, 2014; Karvonen and van Heur, 2014). Studies of urban transitions have emphasised place-specific changes of urban practices and their interactions, while the dynamics of socio-technical systems within and across cities tend to be neglected (Bulkeley et al., 2014; Rohracher and Späth, 2014; Hodson et al., 2016).

The attention to the micro-processes whereby transitions are governed, however, conditions the development of generalizable lessons with wider application in sustainability policy. A recent book, for example, provides an overview of existing theoretical approaches to urban sustainability transitions illustrating the great diversity of approaches that are enriching the field (Frantzeskaki et al., 2017). The book concludes on a cautious note about heterogeneous approaches and calls for integrative frameworks to build understanding across different perspectives. Adapting existing theoretical approaches to the urban context is not sufficient to understand the specific processes that shape the urban transitions (Bulkeley et al., 2010).

This paper integrates different perspectives into a single framework for a spatially-informed perspective on urban energy transitions, the Dimensions of Urban Energy Transitions (DUET) framework. The DUET framework is the culmination of a two-year project on urban transitions in China. This framework emerged from a literature review that analysed both the aspects of urban transitions already reflected in the literature and the blind spots which seemed to be overlooked, or only raised in recent engagements with this problem (Huang and Castán Broto, 2017). Based upon this review, we developed an initial framework to analyse urban transitions- the precursor to DUET- that we used to analyse a qualitative case of urban energy transitions in China (Huang et al., 2018). This helped to refine the framework but raised questions about the extent to which the framework had wider applicability beyond the Chinese context.

We sought to evaluate the applicability beyond the empirical context in which it has been developed, and to consolidate the DUET framework as a theoretical contribution to the sustainability transitions literature. The DUET framework emphasizes different processes involved in transition that may be translatable across context, to allow for an engagement with contextual specificity. Following this we sought to analyse the wider applicability of the DUET framework, and its limitations, through a comparative meta-analysis of a large number of case studies from the literature. This entailed addressing two research objectives: 1) to identify key processes of urban energy transitions applicable in a wide range of case studies; and 2) to conceptualize the key interactions between these processes.

To address the research objectives, the paper proceeds as follows. In Section 2, we introduce the material and methods utilized for the construction of the DUET framework. Section 3 proceeds with a thorough presentation of the framework, the three dimensions of urban energy transitions. Section 4 presents the comparative meta-analysis of 29 cases to further explore the interactions between dimensions of urban energy transitions. Section 5 concludes that efforts to engage with the place-based aspects of transitions can bring broader insights about how transitions actually may happen. The DUET framework provides an approach that takes seriously three dimensions of energy transitions: innovation processes, urban dynamics, and territorial priorities.

2. Material and methods

The objective of this research was to test the DUET framework in relation to a large n-sample of case studies to understand its wider applicability- examining systematically the relevance of the different dimensions in a large sample of cases- and the interactions between different dimensions of the framework reported in those cases. There is a wide literature on sustainability transitions reporting empirical case studies. We used academic databases to search for published case studies of urban transitions using combinations of the following keywords: TOPIC: (urban or city or cities) AND TOPIC: ("sustainability transition*" or socio-technical or "innovation system*" or "energy transition*") AND TOPIC: (energy). The search produced 137 peer-reviewed journal articles of urban energy transitions for the comparative case review. They all included case studies with sufficient information to analyze specific elements in DUET framework. All cases have at its core one of the following six technologies: solar water heaters (SWHs); biogas transport; energy-efficient retrofitting (EER); district heating (DH) and combined heat and power (CHP); electric vehicles (EVs); and natural gas vehicles (NGVs). We categorized the cases into three types: renewable energy technologies in cities; energy-efficient technologies in cities; and transport projects. For each case; we reviewed every dimension of the DUET framework (experimentation; urban governance; territorial socio-spatial arrangements).

To analyze interactions between different dimensions, we selected 29 out of the 137 case studies that documented at least two dimensions of urban energy transitions in the DUET framework (Table 1). The 29 cases studies document 26 cases of transitions in 24 cities. The majority of the cases are from North-western Europe and East Asia (especially China) reflecting the areas that have received greater interest in the literature on sustainable transitions.

Table 1
Selection of key cases of urban energy transitions.

Code	Pioneering energy technology	Case city	Region	Literature
Renewable energy technologies in cities				
RE-1	Solar water heater (SWH)	Dezhou, Rizhao, and Haining (China)	East Asia	Han et al. (2010); Li et al. (2011); Schroeder and Chapman (2014); Goess et al. (2015)
		Mexico City (Mexico)	North America	Mallett (2007)
		Cape town (South Africa)	Africa	Froestad et al. (2015)
RE-2	Biogas transport	Stockholm and Linköping (Sweden), Basel (Switzerland), Odense (Denmark)	North-western Europe	Lienin et al. (2005); Vernay et al. (2013); Fallde and Eklund (2015); Hjalmarsson (2015); Fenton and Kanda (2016)
Energy-efficient technologies in cities				
EE-1	Energy-efficient retrofitting (EER)	Alingsås (Sweden)	North-western Europe	Friesen et al. (2012)
		Ljubljana (Slovenia)	Eastern Europe	Castán Broto (2012)
		Philadelphia (United States)	North America	Edwards and Bulkeley (2015)
		Tianjin (China)	East Asia	Hou et al. (2016)
EE-2	District heating (DH) and combined heat and power (CHP)	Aberdeen, Birmingham, and Woking (United Kingdom); Freiburg (Germany); Stockholm (Sweden)	North-western Europe	Magnusson (2011); Hawkey et al. (2013); Rohracher and Späth (2014); Späth and Rohracher (2015); Webb (2015)
Transport projects				
TP-1	Electric vehicle (EV)	Seoul and Daejeon (South Korea), Shenzhen (China)	East Asia	Kim et al. (2015); Li et al. (2016); Park and Kwon (2016)
		Stockholm (Sweden)	North-western Europe	Olsson et al. (2015)
TP-2	Natural gas vehicle (NGV)	Chongqing, Urumqi, and Beijing (China)	East Asia	Ma et al. (2013); Wang et al. (2015); Hao et al. (2016)
		Delhi (India)	South Asia	Yeh (2007); Krelling and Badami (2016)

3. The Dimensions of Urban Energy Transitions (DUET) framework

Hodson and Marvin (2010a) highlight the mutual constitution of socio-technical regimes interests and urban territorial priorities, and argue for an increasing understanding of how urban governance shapes future priorities. There is a tension between capturing the complexity of the urban milieu in urban infrastructure regimes (Monstadt, 2009) and following the trajectory of a specific technology, to look into the competing or even incompatible relations between alternative technologies that emerge and evolve at the scale of the city. Is there a means to combine a commitment to represent the complexity of socio-technical systems with a clear and transferable analysis of the dynamics of change around specific technologies? For example, renewable energy technologies and energy-efficient technologies are clearly technically and economically feasible nowadays, but their expansion depends on the process that embeds technologies within spatial configurations. Technology change also depends on the conflicts or alignments between the focal industry interests and urban priorities that shape the direction of transitions trajectories.

The DUET framework proposes analysis of urban energy transitions by examining three broad dimensions: 1) the experimental processes leading to the emergence and diffusion of technological innovations; 2) the reproduction of urban processes through context-specific urban politics; and 3) the reconfiguration of socio-spatial arrangements. Each of these dimensions was reviewed in relation to the large set of case studies selected.

3.1. Socio-technical experimentation

The first dimension deals with the diverse, open-ended, but purposive, processes of innovation that are needed to introduce new ideas into a given context, which depend on diverse factors such as the political economy, the material characteristics of infrastructure systems, or the reproduction of institutions and cultural practices. In traditional approaches to transitions theory (e.g. Geels, 2002; Schot and Geels, 2008; Markard et al., 2012), there is a major emphasis on innovation 'niches', protected spaces where experimentation is possible. Socio-technical niches thus emerge within broader technological or sectoral environments that shape experimentation possibilities (Truffer and Coenen, 2012). The DUET framework does not focus on the output (e.g. successful or failed innovations) but rather on the deliberate efforts at fostering processes of experimentation that enable the generation of alternatives.

Technological innovation systems (TIS) scholars emphasise the central role that industry actors often play in conducting pioneering entrepreneurial activities that translate newly-emerging technologies into concrete business opportunities (Negro et al., 2007; Kukk et al., 2016; Planko et al., 2017). In the TIS framework, entrepreneurial activities are regarded as essential for a well-functioning innovation system (Hekkert et al., 2007; Bergek et al., 2008). However, recent work on experimentation and transitions has opened up the door for the recognition of the role that multiple actors may play in sustainability transitions (Bulkeley et al., 2014; Castán Broto and Bulkeley, 2013a, 2013b). First, there are entrepreneurs in all sectors of society. Second, key innovations may be developed for social, rather than business purposes, and generated from processes far away from conventional entrepreneurial or business settings (Nicholls and Murdock, 2012). For instance, Vernay et al. (2013) show how a heterogeneous set of actors engaged in

developing sewage gas for transport in Stockholm, including the municipal government, oil companies, sewage gas refueling stations, and end users. Urban transitions, particularly, open the door to examine socio-technical experimentation as an ambiguous process while also recognizing well-known processes of purposive innovation that relate to conventional understandings of entrepreneurship.

For example, part of our analysis in the DUET framework incorporates the seven functions of the TIS framework (Hekkert et al., 2007; Gosens et al., 2015). Binz et al. (2016) view the maintenance and diffusion of technological innovation as successful mobilization of four key resources: specialized technological knowledge, niche market, financial investment and technology legitimacy. Besides knowledge (technology) and financial resources that are widely recognised as general resources for technological change, legitimacy is also considered as a key resource. The emergence of a niche market for new technology entails alignment between the new products and the place-based regulations, institutions, and social and cultural norms. Legitimacy is often developed through this alignment (Bergek et al., 2008; Binz et al., 2016). Looking at the mobilization of these resources in the 137 cases of urban transitions, we see that they are significant, but that there are also additional social processes of network development that establish the possibility for context-based forms of socio-technical innovation to emerge. Of equal importance is the facilitating role of urban communities and intermediaries in sustainability-oriented experimentation (see for instance: Haughton and Morgan, 2008; Seyfang and Haxeltine, 2012; Seyfang et al., 2014). For example, Edwards and Bulkeley (2015) examine a case of EER in Philadelphia, the project of 'Retrofit Philly "'Coolest Block"' Contest'. The project was based on the knowledge exchange and joint development of the chemical company Dow and the Energy Coordinating Agency. As a key industry player, Dow funded the project (financial investment) and provided technical support for it (knowledge). However, creating suitable innovations in the context of housing retrofitting in Philadelphia required engaging public authorities, and- crucially- the communities of the complexes to be retrofitted. Innovation could not happen without the coordinated action of all those actors that, in relation to the TIS framework, also played a legitimising role for other actors involved in the network.

In sum, innovations may emerge from different forms of entrepreneurship and the mobilization of material and social resources around collective challenges. Entrepreneurs can be thought of not just as those actors who drive business forward, but also, those actors who are willing to bear the risks of a project with unknown results. Therefore, observing the evolving process of new socio-technical configurations requires a broader perspective on the roles of varied actors within and across a specific place.

3.2. Urban political processes

Because of the involvement of a broad range of actors with contesting interests and visions (Späth and Rohracher 2010b; Castán Broto and Bulkeley 2013b), urban energy transitions are inherently political (Truffer and Coenen, 2012). Rutherford and Coutard (2014) emphasize the mutually constituted politics of energy transitions and urban change. Political economy regimes and multi-level governance relationships shape policy strategy directions and the selection of policy instruments even within diverging processes of policy making and implementation (Bulkeley et al., 2014). The dynamics of local political discourses around energy can dramatically change the direction of transitions trajectories (McFarlane and Rutherford, 2008; Späth and Rohracher, 2010a; Hodson and Marvin, 2012; Castán Broto and Bulkeley, 2013a; Moss, 2014).

We think of urban political processes as influencing transitions in three ways: first, the efforts of destabilizing incumbent regimes and the maintenance of the institutional structure of local governance, which tend to secure their own reproduction; second, the active processes to stabilize regimes and to challenge existing states of affairs, that create agonistic spaces for the identification of alternative visions and desires; third, a series of urban material politics that relate to the need to constantly adjust visions of urban change with actual physical transformations in the urban environment. These last material politics include efforts to maintain coherence between discourses and material realities, that may play a role in maintenance of destabilization efforts, but are distinct from them.

The introduction of new technologies into the urban context is often represented as an attempt to destabilize dominant regimes (Bulkeley et al., 2014; Castán Broto and Bulkeley, 2013a, 2013b; Evans and Karvonen, 2014; Evans et al., 2016). However, urban politics may not work in the same way. The operation of different institutions at different levels of government may both initiate radical changes or prevent them (Hodson and Marvin, 2010b; Rutherford and Jaglin, 2015). The introduction of a disruptive technology may be a means to maintain a certain kind of politics at work. For example, the role of local governments in destabilizing dominant regimes is well demonstrated in Shenzhen, where the municipality government took on the task of EV deployment and opened up the local EV market through measures such as providing subsidies and offering free commercial operation license for a given term without fundamentally disrupting the local urban politics and development priorities in the city (Li et al., 2016).

Equally important are agonistic forces that contest sustainability politics, as they may point towards existing contradictions in low carbon policies and move existing or new actors to act (Castán Broto, 2015). In our sample of cases, political contestation between varied actors within and across cities was frequently documented as a significant factor driving transitions. For instance, the application of biogas transport in Stockholm generated disagreement between regional authority and local users, in which the 2010 Regional Development Plan of Stockholm region viewed biogas as the future fuel for vehicles, while a large majority of car owners were not yet ready to compromise with limited fuel availability. Although multiple incentives were provided by the government, they failed to raise the interest of the majority of car owners to switch to sewage gas (Vernay et al., 2013; Hjalmarsson, 2015).

Mediating the above two contesting processes are urban infrastructure networks and built environments (Rutherford, 2014). The mediating role of the urban materiality in urban energy transitions is hard to ignore given a rapidly growing body of literature in transition studies on urban infrastructure (Castán Broto and Bulkeley, 2013a; Hodson and Marvin, 2010a; Moss, 2014) and buildings (Brown and Vergragt, 2008; Castán Broto, 2012). There are complex politics around radical physical alterations of urban environments that involve knowledge production in the business of maintaining urban normality (Bulkeley et al., 2016; Hodson et al., 2016).

For example, a key aspect of urban material politics is the operation of intermediaries in the coordination of different future visions and transformations in urban materiality (Guy et al., 2011).

3.3. Socio-spatial (re)configuration

Pre-existing socio-spatial arrangements pose major challenges to transitions towards a low-carbon energy system (Bridge et al., 2013). New forms of socio-spatial embeddedness and path dependency are also a consequence of the reconfiguration of socio-spatial arrangements in relation to specific energy technologies. We thus approach urban transitions as processes of (re)configuration of 'transition spaces' (Truffer and Coenen, 2012). In particular, the DUET framework emphasizes two aspects of socio-spatial (re) configurations: *territorial proximity* and *socio-spatial embeddedness*. Both mechanisms represent the interplay between evolving socio-technical configuration and local arrangements.

When inquiring into the spatial dimensions of sustainability transitions, Truffer and Coenen (2012) highlight the relevance of proximity. Building upon Boschma's (2005) relational reading of proximity, we conceptualize the proximity for urban energy transitions as the "distance" (in both geographical, operational and cognitive senses) between a new technology and the urban development patterns that can shape the pathways of transitions to a low-carbon energy system. Territorial proximity deals with broader urban contexts that may pose advantages and disadvantages for the emerging niche of certain new energy technology, representing the intense relation of specific local institutional configurations (manufacturing structures, institutional support structures and user profiles) and emergent urban experimentation. This territorial proximity manifests in three main aspects:

1) geographical proximity refers to a locational dimension, e.g. accessibility to resources and clustering of related industries;
2) cognitive proximity reflects the shared knowledge base of actors on certain energy technology, which might differ substantially across cities owing to varied consumer practices;
3) institutional proximity describes the compatibility between the emerging niches and the incumbent urban institutional environment, in terms of both formally determined rules and informally institutionalized cultural norms.

Most reviewed cases confirmed the role of pre-existing proximity of cities. For instance, in the Dezhou case, situating in a bigger industrial belt of geographic concentration of firms, suppliers and associated institutions in solar energy has ensured its advantage in developing and deploying SWH (Li et al., 2011). The development of a solar industrial cluster was regarded as a result of preferential policies, demonstration projects and governmental subsidies. The industry clustering secured SWH dealers with sufficient technical backups and trainings which enabled the anchoring of SWH in local contexts (Han et al., 2010; Li et al., 2011; Schroeder and Chapman, 2014).

Socio-spatial embeddedness depicts the degree to which energy technologies are embedded into urban materialities (e.g. the built environment and the infrastructure fabric) (Bridge et al., 2013). Closely related is the notion of 'obduracy' of cities, especially as regard to urban infrastructures. As described by Hommels (2005), "infrastructures embody heavy investments and capital that are literally sunk in specific locations" (p. 337). Central to this definition is a concern with the physical feasibility of a low-carbon transition and the intangible obstacles for providers and users to shift to low carbon regimes. Urban transitions therefore relate to processes that challenge obduracy (Hommels, 2005; Karvonen and van Heur, 2014).

Territorial proximity and socio-spatial embeddedness simultaneously shape transitions and are shaped by the same processes. On the one hand, because of the material, institutional and cognitive obduracy of incumbent socio-technical systems (Sengers et al., 2016), transitions entail radical transformations of existing place-specific configurations (Monstadt, 2009; Rutherford and Coutard, 2014; Bulkeley et al., 2014). On the other hand, both proximity and embeddedness are rearranged and reconfigured around new technologies, during which process accessibility to resources might be improved, knowledge developed and institutions adapted (Truffer and Coenen, 2012). The DUET framework is a tool to go beyond analyses of transitions *in* cities to focus, instead, in transitions *of* cities.

3.4. An integrated analytical framework

The Dimensions of Urban Energy Transitions (DUET) framework brings together three dimensions of change (socio-technical experimentation, urban politics, and socio-spatial transformations) not as entirely different spheres of change, but rather, as simultaneous processes that may reinforce or weaken each other. The DUET framework differentiates efforts to produce newness (experimentation) and gaining authority (politics) within a given spatial context (reconfiguration).

Fig. 1 summarizes the three dimensions of the DUET framework. The mobilization of four system resources represents practices of maintaining and upscaling socio-technical experimentation. Urban political processes are reflected in three key components of destabilizing dominant regimes, stabilizing incumbent regimes, and urban material politics. The (re)configuration of socio-spatial arrangements reflects the emergence and stability of new patterns of territorial proximity, along with processes of disruption of the incumbent regime and the formation of new socio-spatial embeddedness. Materiality is a focal area, and actors might simultaneously take part in socio-technical experimentation and urban politics. Fig. 1 depicts the overlapping of urban processes and energy transitions that encompasses both emergence and diffusion of socio-technical experimentation and (re)configuration of socio-spatial arrangements. The three foci enable analyzing the multiple dimensions of urban energy transitions in the DUET framework. In next section, a meta-analysis is systematically presented to reveal the three key interactive processes in real-world cases.

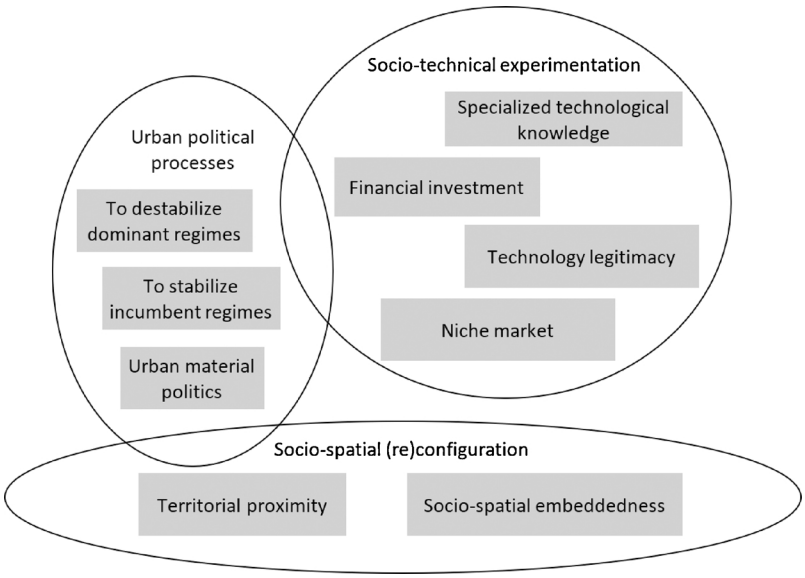

Fig. 1. The three dimensions of urban energy transitions: the DUET framework.

4. Interdependence of urban processes and energy transitions

This section addresses the need to understand the interactions between the different dimensions of the framework, to support a full analysis of the spatial aspects of urban energy transitions. Based on the comparative meta-analysis of 29 cases, we identify three major types of interactions: interactions between urban political processes and socio-technical experimentation, the impact of pre-existing socio-spatial arrangements on the experimentation and corresponding urban changes, and the aligning of urban priorities and new technologies that reconfigure socio-spatial arrangements.

4.1. Interactions between urban political processes and socio-technical experimentation

Innovation is a collective activity deployed through the complex interactions among a multitude of different actors and organizations, that are enabled and constrained by physical artifacts and by institutions that pose 'the rules of the game' (Carlsson and Stankiewicz, 1991; Hekkert et al., 2007; Bergek et al., 2008). The mutual influence between urban processes and socio-technical experimentation implies that experimentation of new technologies directly interacts with urban actors, but has less direct impact on actual transformations of the urban fabric. The case review shows that, of the three urban political processes, the destabilization of urban regimes (often through local governmental governance) plays a definitive role in shaping socio-technical change in cities. The power and capacity of territorial interests determine socio-technical reconfiguration. Industrial interests in cities seem to have influence particularly when they are aligned to urban priorities.

In the case of EV in Shenzhen, both the local government and major business actors fostered the socio-technical change of EV. On the one hand, the Shenzhen municipality signed a franchising agreement to approve Potevio – a state-owned enterprise – to act as the licensed charging network supplier and E-bus-related intermediary business operator. This agreement enabled Potevio easy access to long-term loans from banks. On the other hand, as a licensed operator for the charging infrastructure network in Shenzhen, Potevio offered the Shenzhen Bus Group an eight-year lease for EV bodies, which significantly reduced the financial pressure on the Shenzhen Bus Group in purchasing E-buses. Moreover, Potevio separated the leasing of batteries from the bodies. In other words, Potevio kept the battery ownership and took the responsibility to maintain and renew the batteries, which again reduced the high battery-related costs of the Shenzhen Bus Group (Li et al., 2016). As a result, Shenzhen is recognised worldwide as a successful case for E-buses transitions with a total of 16359 E-buses operating till 2017.[1] In the case of EV and Biogas transport in Stockholm, public procurement and subsidies acted as significant impetus for niche market formation (Vernay et al., 2013; Olsson et al., 2015). Similarly, the case of EER in Ljubljana documented the impact of urban governance on knowledge development and diffusion. The Ljubljana municipality built partnerships with local academic institutions such as the Building and Civil Engineering Institute of Ljubljana and the Faculty of Civil and Geodesic Engineering, enabling the development and diffusion of EER knowledge across energy and construction sectors (Castán Broto, 2012).

The case of Biogas transport in Odense demonstrates the importance of processes of conflict and contestation. The municipality rejected biogas in favour of diesel hybrid bus technology in a 2014 procurement of new buses, which served as part of the reason for the failure of niche market formation and the fade out of biogas transport (Fenton and Kanda, 2016).

[1] http://www.sztb.gov.cn/xxgk/jtzx/gzdt/gjdt_80994/201712/t20171228_10630454.htm.

Turning to urban politics that often shape the discourses around energy transitions, in the case of EER in Alingsås, the regular participative meetings with the residents and general contractor organized by local government enabled positive discourses on the renovation project, through which the residents were well informed, pointed questions, and took advantage of their involvement, leading to rather positive attitudes toward these projects (Friesen et al., 2012). The development of SWH in Mexico City showed the institutional difference between vested urban interests and emerging needs for change. And the Petroleos Mexicanos (PEMEX), as the state-run petroleum industry, seemed to act as an obstacle for the deployment of SWH (Mallett, 2007).

4.2. Pre-existing socio-spatial arrangements that promote or inhibit the alignment of socio-technical experimentation and urban political processes

Territorial proximity, especially institutional proximity, seems to have a stronger influence on urban transitions than socio-spatial embeddedness. Our analysis confirms Truffer and Coenen's (2012) theoretical observation that there is a need to take into account the social, spatial, and institutional configuration in transition studies. The key role of pre-existing socio-spatial arrangements also demonstrates the power and capacity of incumbent urban priorities and their interests in promoting or prohibiting the emergence and development of socio-technical experimentation in cities. It seems that successful socio-technical transitions in cities will only be possible if the industry interests are aligning with urban priorities, especially at the early stage of experiments.

Territorial proximity poses significant impacts on urban political processes and the diffusion of socio-technical experimentation. With respect to the impact of geographical proximity on the diffusion of experimentation, the case of NGV in Chongqing demonstrated that the concentration of NGV vehicles, which were mainly operated by public bus companies, was to a large extent supported by richness in natural gas resources (Ma et al., 2013). High logistics costs inhibited the scaling up of biogas production in Odense, because of the large distance between urban areas and farm-based biogas production sites (Fenton and Kanda, 2016). In the case of Dezhou and Chongqing, spatial proximity to related industrial clustering facilitated the success of transition. In the case of SWH in Dezhou, the development of a solar industrial cluster secured SWH dealers with sufficient technical backups and trainings, which resulted in good and reliable maintenance service for end-users and further enabled the expansion of local markets (Li et al., 2011). In the case of NGV in Chongqing, the clustering of CNG manufacturing industry substantially lowered the purchase and operating costs of the local CNG vehicles, which generated an occupation of 30% of the Chinese market (Ma et al., 2013).

Institutional proximity also shapes the governance capacity of different urban actors, as well as innovation's legitimacy. In the case of SWH in Mexico City, there was a mass boycott of the emerging SWH technology. The concept of making such a large investment in SWH in which payback may be seen in about five years was incongruent with a culture in which many people tended to live day by day. As a result, the social acceptance of SWH remained low (Mallett, 2007). In contrast, in the case of SWH in Dezhou, a pre-existing adaptive and resilient attitude towards resource scarcity among Dezhou's residents helped them to modify habits and, being aware of the inexhaustible attribute of solar energy, to adopt new technologies like SWH (Li et al., 2011).

Pre-existing socio-spatial configurations seem to act as an irreversible element in socio-technical transitions in cities, which easily shape the emerging socio-technical niche and it is hardly moveable merely through experimentation. In the case of Biogas transport in Odense, making use of existing urban infrastructure networks, the NGF Nature Energy company combined the provision of biogas with existing natural gas grid and fuel stations, and injected biogas into the natural gas grid (Fenton and Kanda, 2016).

4.3. Alignments of urban political processes and socio-technical experimentation that reconfigure socio-spatial arrangements

Reconfiguration of socio-spatial arrangements implies the anchoring of emerging socio-technical institutional configurations and the formation of new path-dependency and lock-in. With regard to energy technologies, only a few cities demonstrated signs of new socio-technical configurations. Introduction of new technologies always acts as the starting point of the whole process from transitions in cities to transitions of cities. A broader reach is needed for the anchoring of new socio-technical institutional configurations in cities. Overall, broad socio-technical changes will occur only when urban priorities align with industry interests, along with powerful urban actors, especially urban governments, taking over the task of transitions.

In the case of SWH in Dezhou, the solar industrial agglomeration revolved around the biggest SWH manufacturer in the world – Himin – which acted as the system-builder in developing the whole supply chain of SWH. As a result, more than 100 solar industry related companies registered in Dezhou. In addition, three out of every ten jobs in Dezhou are solar-related, making solar industries the key pillar of the local economy and employment. This wider involvement with solar industry inevitably shaped households' perceptions of green technologies and changed their consumption custom. For instance, Dezhou's households purchased various solar-related products, ranging from low-cost solar toys and solar electrical appliances to high-cost products such as SWHs (Li et al., 2011). In the case of Biogas transport in Stockholm, a local company undertook a heavy investment in 130 biogas buses, a filling station, and a pipeline to connect the upgrading plant to one of its bus depots, which represented a large volume of sunk costs, creating irreversibility regarding the enrolment of biogas buses in the biogas network (Vernay et al., 2013).

Except for the expansion of local markets, knowledge diffusion of emerging technologies will also lead to the formation of institutional and cognitive proximity. In the case of EER in Alingsås, a public housing company – Alingsåshem – together with the Tenants Society established a local newsletter that enabled the alignment of different perceptions on EER projects. For the general contractor and Alingsåshem, it was a successful forum to share detailed updates on the renovation process. For the Tenants Society it was a platform to advertise formal meetings, social community events, as well as to address residents' questions and concerns. For residents, the newsletter provided renovation and passive house education in a simple and tangible style (Friesen et al., 2012). Besides, individual practices are able to create proximity. In the case of DH and CHP in Aberdeen, based on access to formal and

Table 2
Interdependence of urban processes and energy transitions of the six pioneering energy technologies in cities.

Code	Pioneering energy technologies in cities	Total number of case cities	❶ Interactions between urban political processes and socio-technical experimentation	❷ Impacts of pre-existing socio-spatial arrangements on the interactions	❸ Impacts of the interactions on socio-spatial (re) configurations
RE-1	SWH	5	+ + + +	+ + +	+ +
RE-2	Biogas transport	4	+ + + +	+ +	+
EE-1	EER	4	+ + + +	+ + +	+
EE-2	DH and CHP	5	+ + + +	+ +	+
TP-1	EV	4	+ + + +	NA	NA
TP-2	NGV	4	+ +	+ +	NA

Note: Each ‘ + ’ denotes a case city that has been documented in the literature.

informal social networks, the experience in anti-poverty campaigns, negotiation skills and personal influence, one officer’s lobby activities managed to increase social acceptance of DH and CHP (Hawkey et al., 2013).

To sum up, the DUET framework is a means to examine systematically the multiple interrelated processes that lead to an urban energy transition. Interactions between different dimensions tend to create obdurate mechanisms which reinforce incumbent technologies. This obduracy determines its resistance to efforts aimed at changing it (Hommels, 2005). In line with previous transition research, forging alignment is crucial to foster a transition (Truffer and Coenen, 2012). However, this is not just an alignment of institutions and organisations, but also an alignment of spatial patterns and innovation policies. Table 2 presents results of the meta-analysis that explores the interactions between socio-technical experimentation, urban political processes, and socio-spatial (re) configuration for each technology. Each ‘+’ denotes a case city in the literature that has captured the corresponding interactions. Because each dimension encompasses sub-dimensions, if there exhibit any interactions between sub-dimensions, the corresponding city would be counted as one case. The analysis suggests that there are multiple interactions between the different elements of the DUET framework. Yet, there is not enough alignment – e.g. coordination, common governance, planning − to catalyse broader socio-spatial reconfigurations of the kind envisaged in energy transitions.

The DUET framework enables the analysis of specific patterns of interaction. The introduction of a new technology may induce multiple urban political processes, while urban politics driven by different interests and visions may influence the outcome of socio-technical experimentation (process ❶ in Fig. 2). Pre-existing socio-spatial arrangements may act as contextual factors that shape both urban political processes and socio-technical experimentation, as well as their interactions (process ❷ in Fig. 2). Socio-spatial arrangements may be transformed in turn through continuous, sustained and in many cases long-lasting interactions between experimentation and urban processes (process ❸ in Fig. 2). The meta-analysis of case studies in Table 2 suggests that, process ❶ exhibits the strongest interaction, followed by process ❷. While socio-spatial reconfiguration (process ❸) is somewhat elusive. This implies that ‘transitions of cities’ entail even deeper embedding of the technology into urban systems, which requires tremendous, continuous, and coordinated efforts of manifold actors.

The DUET framework marks a change of thinking from viewing urban energy transitions as a process of ‘transitions in cities’ to focus in ‘transitions of cities’. The framework does not propose an additive approach to consider urban process and energy transitions simultaneously. Rather, it acknowledges the mutual dependence of both processes. The DUET framework moves away from a unilateral understanding of city as a “container” with homogeneous contexts (as criticised in Bulkeley et al., 2010; Coenen and Truffer, 2012; Hodson et al., 2016). The role of city in energy transitions is dynamic. In ‘transitions of cities’, urban priorities play a key role in the strategic actions of dominant actors such as governmental or business organisations. Urban actors act as intermediaries who resolve conflicts or facilitate the alignment between industry interests and territorial priorities (Moss, 2012).

Conflict and alignment of industry interests and urban priorities co-emerge in urban transitions towards sustainability. There is a feature, for example, of conflicts related to large-scale on-shore wind farm and land use (Felber and Stoeglehner, 2014). Alignment is needed to manage trade-offs between different technologies, such as in the case of photovoltaic systems installation and agricultural practices, since ground-mounted PV plants are often depicted in competition with crop production for agricultural land (Sacchelli et al., 2016). These conflicts or alignments at the scale of the city reveal the interdependence between the practices of industry actors

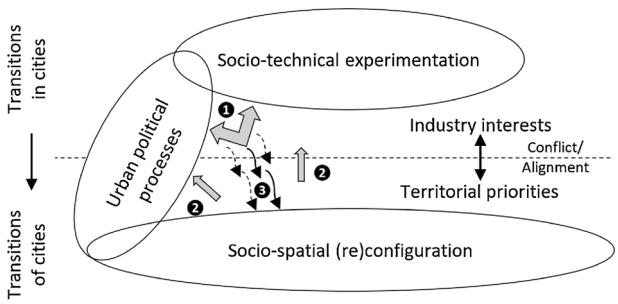

Fig. 2. DUET framework to study the interdependence between urban processes and energy transitions.

in cities and urban governance priorities. Transition possibilities will depend on a variety of factors such as the development of mutual relationships, the possibility of technical, spatial, or economic contiguity, or the establishment of competing future visions. Our analysis of existing case studies suggests that, in practice, the role of industry interests is often ignored. However, the cases of SWH and NGV transitions in several cities suggest that industry clustering promotes transition processes. Hence, that clustering cannot be explained by the agglomeration of industrial interests alone. In urban settings, there is a change from transitions which tend to follow business and industry development priorities at the national level to transitions that follow urban priorities (whether this is explicit in strategic planning instruments or not). Such finding points towards the need to bridge understandings of socio-technical change in cities and existing analyses of socio-spatial reconfigurations of cities. This is not anymore a question of recognising urban priorities, but of ensuring that urban transition strategies and industrial policies are well coupled.

5. Conclusion

Overall, the DUET framework provides a basis for the examination of the different forms of interaction between processes of socio-technical experimentation, urban governance, and territorial control. A key contribution of this framework is that it combines established views of socio-technical transitions with recent advances in the emerging field of urban energy transitions, and responds to the calls for more nuanced insights into spatial dynamics in sustainability transitions (Rutherford and Coutard, 2014; Truffer and Coenen, 2012). The DUET framework bridges theories of energy transitions study with the geography of transitions. In doing so, it represents the complexity of the dynamics of change within a socio-technical system with the depth of place-based understanding of the dynamics of transitions. This dynamic is particularly visible in the rapidly growing cities of the urbanisation age.

The comparative analysis shows that urban governance factors play a decisive role in sparkling multiple forms of experimentation in cities. This underlines the role of local governments in transitions. In contrast, processes of innovation have limited influence in the overall processes of urban transformation that the city is embedded within. This highlights that we may have a better idea about how to change socio-technical systems (defined around one technology) than how to change cities. This is not a new question for urban scholars, but it opens up a set of interesting questions for the transition community.

In our sample, pre-existing socio-spatial configurations appear to act as an irreversibility element in socio-technical transitions in cities. Our cases confirm that, socio-spatial reconfiguration occurs through long and arduous processes, which requires continuous and collective practices of both experimentation and urban governance. A broad socio-technical change will occur only when urban priorities align with industry interests, along with powerful urban actors, especially urban governments, taking over the task of transitions. The DUET framework provides a means of analysis to structure information to understand the complex interactions between urban processes and energy transitions.

The DUET framework responds to the criticisms of the overly mechanistic framing of transitional processes toward sustainability (Truffer and Coenen, 2012). The TIS framework, for example, assumes that the formation of a new socio-technical system implies a fundamental societal change toward sustainability. By emphasizing the political and spatial aspects of transitional process, the DUET framework points towards the complexity of social change. We encapsulate this idea in the slogan to move away from “transitions in cities” to “transitions of cities”. For example, what happens when socio-technical change succeeds/fails to reshape social practices of citizens (implying a more sustainable way of energy consumption and everyday life) and the built environment of a city (implying a lock-in of a more sustainable and low-carbon society).

Moreover, the DUET framework focuses on simultaneous processes of transition. For example, the DUET framework can be used to analyse complexities between competing technologies in a given city, which differentiates it from conventional frameworks such as multi-level perspective (MLP) and TIS. By explicitly taking into account the political and spatial dimensions, the interactions between different urban actors that represent competing technologies are observable. Moreover, the competing relations can be analysed by looking into the interactions between the three dimensions of transitions. For example, the practices to destabilize other regimes for cumulating varied resources for certain technology (e.g. lobbying for certain strategy that excludes other technology solutions) imply less resources for competing technologies. Comparative methodologies may help to explore this question empirically.

Our work, so far, also depends on the dominant geographical realms of transition studies. There has been a bias in the transitions literature towards cases in urban areas with strong local governments (such as in Western Europe) or in cases where local governments have been vested with new powers to act for an energy transition (e.g. China). There is however a pressing need to understand transitions in a wider range of cities, including secondary cities and cities who do not fit squarely within international flows of technology and ideas. There is also a need to understand urban transitions in context in which local governments lack a mandate to implement sustainability policies. In the context of rapid urbanisation, researchers of urban transitions need to transcend the familiar sites of sustainability action.

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